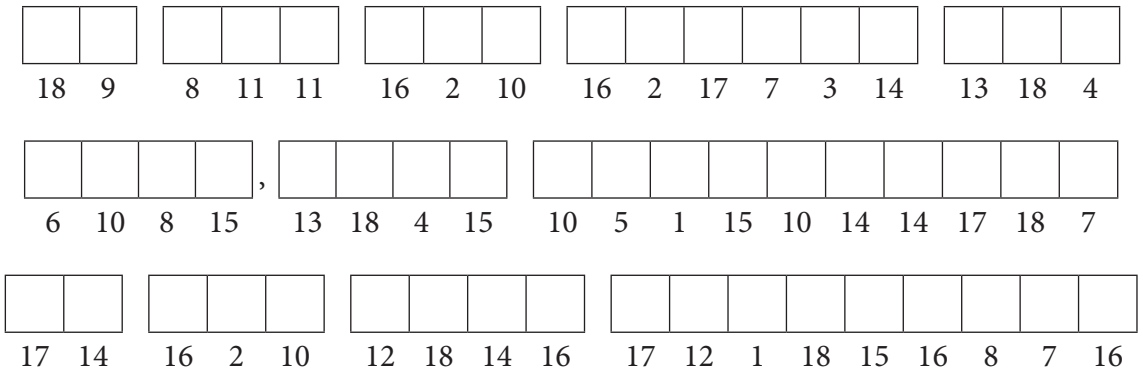
Name: _____

WORKSHEET

Geometric series



Janet Lane

Rule

The sum of a geometric progression is called a geometric series. The sum of the first n terms in a geometric progression with first term a is:

$$S_n = \frac{a(1-r^n)}{1-r} \text{ where } r < 1$$

or

$$S_n = \frac{a(r^n-1)}{r-1} \text{ where } r > 1.$$

Questions

Calculate the sum of the following geometric progressions.

1 The first 5 terms of 7, 21, 63, …

2 The first 11 terms of 5600, 2800, 1400, …

3 The first 8 terms of 2, −8, 32, …

4 The first 20 terms of $\frac{3}{4}, \frac{3}{5}, \frac{12}{25}, \ldots$

5 The first 30 terms of $\frac{x}{693}, \frac{2x}{693}, \frac{4x}{639}, \ldots$

6 $3 + 6 + 12 + \ldots + 3072$

7 $\frac{7}{2} + \frac{7}{5} + \frac{14}{25} + \ldots + \frac{56}{625}$

8 $4 - 4.04 + 4.0804 - \ldots - 4.288\ 54$

Solutions

A $-0.164\ 89$

E $\frac{5(y^9 - 6561)}{2187y^7(y-3)}$

F 514.985

G $-26\ 214$

H $11\ 194.531\ 25$

I $\frac{x^{10} - 512x}{x-2}$

L 15

M 13

N $5\frac{967}{1250}$

O 1488.32 kg

© Cengage Learning Australia Pty Ltd 2013

9 $2 + 3 + 4.5 + \ldots + 172.995$

10 $\frac{5}{y^7} + \frac{5}{3y^6} + \frac{5}{9y^5} + \ldots + \frac{5y}{6561}$

11 How many terms of the series $\frac{2}{3} + 2 + 6 + \ldots$ need to be added together to equal $4\,782\,968\frac{2}{3}$?

12 Find n if $S_n = 23\,488\,102\frac{3}{4}$ for $\frac{7}{4} - 7 + 28 + \ldots$

13 Find n if $S_n = 430$ for $10 - 20 + 40 - \ldots$

14 A sum of \$200 is invested in a travel fund and each month the traveller invests 10% more than the month before. How many months will it take the traveller to save for a \$12 800.50 trip through South America?

15 If the 3rd term of a geometric sequence is 486 and the 7th term is 6, find the sum of 8 terms.

16 For a geometric series, find S_{11} if $t_2 = -\frac{3}{4}$ and $t_5 = -6$.

17 For a geometric series, find S_9 if $t_4 = 32x^4$ and $t_7 = 4x^7$.

18 A first-time fruit picker is able to increase the amount he picks each day by 20% over his first week. If he picked 288 kg on the third day, what was the total amount he picked from Monday to Friday?

P 847

R 6560

S 21

T $-767\frac{5}{8}$

U 3.7068

W 6141

X 1 549 411x

Y 7

© Cengage Learning Australia Pty Ltd 2013